Example 3.55 Solve $\sqrt{3}\tan^2\theta + \left(\sqrt{3}-1\right)\tan\theta - 1 = 0$

Solution:

$$\sqrt{3}\tan^2\theta + \left(\sqrt{3}-1\right)\tan\theta - 1 = 0$$
$$\sqrt{3}\tan^2\theta + \sqrt{3}\tan\theta - \tan\theta - 1 = 0$$
$$\left(\sqrt{3}\tan\theta - 1\right)(\tan\theta + 1) = 0$$

Thus, either $\sqrt{3}\tan\theta - 1 = 0$ (or) $\tan\theta + 1 = 0$

If $\sqrt{3}\tan\theta - 1 = 0$, then	If $\tan\theta + 1 = 0$ then
$\tan\theta = \frac{1}{\sqrt{3}} = \tan\frac{\pi}{6}$	$\tan\theta = -1 = \tan\left(\frac{-\pi}{4}\right)$
$\Rightarrow \theta = n\pi + \frac{\pi}{6}, n \in \mathbb{Z}$...(i)	$\Rightarrow \theta = n\pi - \frac{\pi}{4}, n \in \mathbb{Z}$...(ii)

From (i) and (ii) we have the general solution.

Exercise - 3.8

1. Find the principal solution and general solutions of the following:
   (i) $\sin\theta = -\frac{1}{\sqrt{2}}$ (ii) $\cot\theta = \sqrt{3}$ (iii) $\tan\theta = -\frac{1}{\sqrt{3}}$.
2. Solve the following equations for which solutions lies in the interval $0^\circ \leq \theta < 360^\circ$
   (i) $\sin^4 x = \sin^2 x$
   (ii) $2\cos^2 x + 1 = -3\cos x$
   (iii) $2\sin^2 x + 1 = 3\sin x$
   (iv) $\cos 2x = 1 - 3\sin x$.
3. Solve the following equations:
   (i) $\sin 5x - \sin x = \cos 3x$
   (ii) $2\cos^2\theta + 3\sin\theta - 3 = 0$
   (iii) $\cos\theta + \cos 3\theta = 2\cos 2\theta$
   (iv) $\sin\theta + \sin 3\theta + \sin 5\theta = 0$
   (v) $\sin 2\theta - \cos 2\theta - \sin\theta + \cos\theta = 0$
   (vi) $\sin\theta + \cos\theta = \sqrt{2}$
   (vii) $\sin\theta + \sqrt{3}\cos\theta = 1$
   (viii) $\cot\theta + \text{cosec}\theta = \sqrt{3}$
   (ix) $\tan\theta + \tan\left(\theta + \frac{\pi}{3}\right) + \tan\left(\theta + \frac{2\pi}{3}\right) = \sqrt{3}$
   (x) $\cos 2\theta = \frac{\sqrt{5}+1}{4}$
   (xi) $2\cos^2 x - 7\cos x + 3 = 0$